

Voluntary Report – Voluntary - Public Distribution

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Report Name: Vietnam Biofuels

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Post: Hanoi

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Prepared By: Nguyen Linh

Approved By: Benjamin Henderson

Report Highlights:

Starting June 1, 2026, Vietnam took a significant step in its energy transition by requiring bioethanol blending in all gasoline grades, meaning pure fossil gasoline is no longer commercially available. Biodiesel blending continues to be voluntary for now. To keep up with growing demand, Vietnam is ramping up both domestic bioethanol production and imports.

SECTION I: EXECUTIVE SUMMARY

Vietnam's biofuel sector continues to develop gradually, driven by government mandates and climate commitments, though significant challenges remain including infrastructure, logistics, and consumer acceptance. The country committed to achieving net-zero emissions by 2050 at COP26 and is working to update its Nationally Determined Contribution (NDC) submitted to the UNFCCC in October 2022. The Ministry of Industry and Trade (MOIT) leads biofuel policy implementation in Vietnam. In November 2025, MOIT issued Circular 50, establishing a new blending roadmap requiring all gasoline sold nationwide to contain ethanol starting June 1, 2026. Under this policy, RON92 gasoline may continue with 5 percent ethanol (E5) through 2030, while all other gasoline grades must contain 10 percent ethanol (E10). Post forecasts Vietnam's fuel ethanol imports will rise more than forty-fold to 490 million liters in 2026. Imports are up on limited production growth and greater fuel ethanol consumption, driven by two key policy changes that took effect June 1: the nationwide E10 blending mandate and the elimination of pure fossil gasoline from the market.

Vietnam's E5 mandate, introduced in 2018, required fuel distributors to offer E5 gasoline at retail stations nationwide. Implementation has faced challenges including consumer resistance and a lack of clear pricing incentives. Fuel ethanol consumption in 2025 is estimated at 80 million liters. Vietnam has seven ethanol production facilities, though only four currently operate, with a combined nameplate capacity of approximately 320 million liters per year. Domestic production remains insufficient to meet demand, necessitating continued imports. The United States has been the main supplier in recent years. Production challenges include volatile cassava prices, reliance on imported corn as feedstock, and less efficient conversion technologies.

Vietnam's transportation sector is transitioning toward electric vehicles, supported by tax and fee incentives. Special consumption tax rates for battery electric automobiles are 1–3 percent through March 2027, compared to 10–35 percent for fuel-use automobiles, while registration fees for electric vehicles are zero percent through December 2030. Total gasoline consumption in 2025 is estimated at 10.5 billion liters, while diesel consumption reached 12.5 billion liters. The growth of electric vehicles and the expansion of metro systems in Hanoi and Ho Chi Minh City may moderate future growth in the fuel pool, but won't have an immediate impact, as gasoline consumption is forecast to grow 2.6 percent in 2026.

Biofuel trade is open to businesses in Vietnam. Denatured ethanol faces a 5 percent MFN tariff but enters duty-free under free trade agreements, while biodiesel enters duty-free under most trade agreements.

Biodiesel production and consumption are minimal in Vietnam, due to no nationwide biodiesel mandate in Vietnam, economic viability challenges, and tight feedstock supplies.

Vietnam does not currently produce sustainable aviation fuel (SAF). In August 2025, MOIT issued Directive 13 on SAF development and use for aviation transportation, assigning responsibilities to develop domestic SAF production from agricultural feedstocks including bagasse, cassava peels, rice husks and straw, corn husks and cobs, sugarcane bagasse and molasses, used cooking oil (UCO), and other biomasses. Vietnam Airlines and VietJet Air have conducted initial SAF pilot flights using imported SAF, while Petrovietnam is conducting feasibility studies for potential biorefinery projects.

ACRONYMS AND ABBREVIATIONS

BAV	Battery Electric Automobile
HV	Hybrid Vehicles
NDC	Nationally Determined Contribution
UNFCC	United Nations Framework Convention on Climate Change
UCO	Used Cooking Oil
MAE	Ministry of Agriculture and Environment
MOIT	Ministry of Trade and Industry
MOF	Ministry of Finance
MOC	Ministry of Construction
MOST	Ministry of Science and Technology
MFN	Most Favored Nation
NTR	National Technical Regulation
IATA	International Air Transport Association
JETP	Just Energy Transition Partnership
TOE	Tons of Oil Equivalent
PVN	Petrol Vietnam – Vietnam National Industry – Energy Group
PV Oil	Petrol Vietnam Oil Corporation
Petrolimex	Vietnam National Petroleum Group
VAMA	Vietnam Automobile Manufacturers' Association
VAMM	Vietnam Association of Motorcycle Manufacturers

SECTION II: POLICY AND PROGRAMS

A. Renewable Energy and Greenhouse Gas Emissions

Vietnam submitted its Nationally Determined Contribution (NDC) to the UNFCCC in October 2022 and is currently working on NDC updates. At the COP26 climate summit in 2021, Vietnam committed to achieving net-zero emissions by 2050, a pledge that increased attention on renewable energy development, including biofuels. The Just Energy Transition Partnership (JETP), announced in 2022 with international support, provides potential funding for renewable energy projects, though biofuels were not explicitly prioritized in initial JETP allocations. Currently, there are currently no laws requiring biofuels consumed in Vietnam to meet specific GHG reduction thresholds, and there are no laws influencing disallowing or creating a preference for a particular feedstock.

Vietnam's National Green Growth Strategy (2021–2030) aligns with the country's Nationally Determined Contribution (NDC) under the Paris Agreement, which commits Vietnam to reducing greenhouse gas emissions intensity nine percent by 2030. The strategy identifies renewable energy, including biofuels, as a key component of Vietnam's decarbonization pathway. However, the strategy lacks specific biofuel production targets or mandatory blend levels beyond the existing E5 mandate. Implementation responsibility is distributed across multiple ministries, with MOIT leading on fuel standards and distribution and MAE overseeing feedstock development and environmental compliance.

The Revised Law on Environmental Protection (2020) provides the legal foundation for environmental standards, including emissions controls that indirectly support biofuel adoption. The law requires environmental impact assessments for biofuel production facilities and establishes frameworks for waste management, including UCO collection.

B. Policy and Programs on Biofuels

Policy Framework

The Ministry of Industry and Trade (MOIT) leads biofuel roadmap development and implementation in Vietnam. MOIT issued Circular 50, dated November 11, 2025, to regulate the blending roadmap of biofuels with conventional fossil fuels in Vietnam. Starting June 1, 2026, all gasoline sold nationwide must be blended with ethanol. Under this policy, 92-octane gasoline may continue to be sold with a 5 percent ethanol blend (E5) until the end of 2030, while all other gasoline grades must contain 10 percent ethanol (E10). Three departments under MOIT are responsible for implementing the biofuel roadmap. The Agency for Innovation, Green Transition and Industry Promotion (IGIP) is responsible for technical standards and regulations for biofuel and ethanol gasoline production, storage, blending, transportation, and distribution. The Agency for Domestic Market Surveillance and Development (ADMS) is responsible for guidance and monitoring of biofuel and ethanol gasoline imports, distribution, and trade, and collaborates with the Ministry of Finance to propose tax and fee policies to promote biofuel use. The Department of Petroleum and Coal (DPC) is responsible for promoting domestic biofuel production.

1. Fuel Ethanol and Gasoline Market

Table 1. Bioethanol Blends: Date Piloted, Mandated, and Implemented

Blend	Piloted	Date Mandated and Policy	Date Implemented
E5 RON92	August 01, 2010, in seven provinces.	Prime Minister Announcement 255, June 6, 2017.	01 January 2018 (phased out on December 31, 2030)
E10 for RON95 and other gas grades	August 01, 2025, in three provinces	MOIT Circular 50, November 7, 2025.	June 01, 2026

Source: Post analysis based on cited legal documents.

Blend Mandate

Vietnam's E5 gasoline mandate, introduced in 2018 following several years of pilot programs, requires fuel distributors to offer E5 at retail stations nationwide. The mandate was designed to reduce dependence on imported gasoline, support domestic agriculture through feedstock demand, and reduce vehicle emissions in urban areas. Since January 1, 2018, Vietnam has officially maintained E5 RON92 while continuing to allow pure fossil fuel RON95.

Vietnam regulated E10 as a voluntary mechanism since introducing E5 as a pilot and mandatory blend. However, E10 was not available in the market until it was mandated under MOIT Circular 50, in November 2025, with mandatory implementation beginning June 1, 2026, after which pure fossil fuel gasoline is no longer permitted. Vietnam currently offers both E5 and E10 gasoline grades in the market.

Implementation of the E5 mandate has faced challenges.

Consumer Acceptance: Initial consumer resistance to E5 gasoline has gradually decreased, though concerns about engine compatibility and fuel efficiency persist.

Pricing Mechanisms: The lack of clear pricing incentives for E5 versus conventional gasoline has led to a narrow price gap that has limited consumer uptake in areas where both fuels are available.

Fuel Quality Standards

The Ministry of Science and Technology (MOST) issued Circular 16 in 2022 and Circular 19 in 2026, regulating bioethanol standards and aligning HS codes as shown in the table below. These are the National Technical Regulation (NTR) 01/2022/BKHCN and its amendment, NTR 01:2026/BKHCN, on gasoline, diesel fuel oils, and biofuels. Vietnam is drafting an additional NTR to align with the E10 nationwide implementation roadmap, however definition of denatured and undenatured fuel ethanol are not required to produce from biological feedstock like the current NTR. This may allow Vietnam to source different types of fuel ethanol including synthetic ethanol.

Table 2: Bioethanol Quality Standards

Blending	Document	Issued and effected date
E5, E10 and E100	MOST Circular 16 issued on December 15, 2022	June 15, 2023
	MOST Circular 19 issued on April 30, 2026	April 30, 2026 Oxygen content criteria are valid till December 31, 2026

Table 3: HS Codes

Product description	HS code
Non-lead gasolines	2710.12.21; 2710.12.23; 2710.12.24; 2710.12.26
Ethanol blended gasolines	2710.12.22; 2710.12.25
Denatured ethanol	22.07.20.11; 2207.20.19; 2207.20.90
Undenatured ethanol	2207.10.00

Source: Post analysis based on cited legal documents.

2. Biodiesel/Renewable Diesel and Diesel Market

Blend Mandate

Unlike ethanol, Vietnam has not implemented a nationwide biodiesel mandate. B5 and B10 diesels are voluntary. Biodiesel development remains limited to pilot programs and small-scale production facilities. The following factors have constrained biodiesel sector development.

Limited Feedstock Base: Vietnam produces limited quantities of oilseeds suitable for biodiesel production. Vietnam imports soybeans for crushing for soy oil and imports palm oil for consumption, food service and processing. UCO collection is low in Vietnam. According to local industry, in recent years they have done the UCO collection on a small scale, then conducted semi processing for exporting.

Economic Viability Concerns: B5 biodiesel (five percent blend) has been tested in select provinces, including Ho Chi Minh City, Binh Duong, and Dong Nai, primarily in captive fleets such as buses, trucks, and government vehicles. Results have been mixed, with technical performance generally acceptable but economic viability questionable due to high feedstock costs.

Regulatory Uncertainty: The absence of a national biodiesel mandate creates uncertainty for potential investors. Without guaranteed demand through blending requirements, private sector investment in biodiesel production capacity remains minimal.

Fuel Quality Standards

The National Technical Regulation (NTR) 01/2022/BKHCN on gasoline, diesel fuel oils, and biofuels provides technical quality regulations for diesel fuel, biodiesel fuel, and B100.

Table 4: Fuel HS codes

Product description	HS code
Fuel diesel	2710.19.71; 2710.19.72
Biodiesel blended fuel	2710.20.00
B100	3826.00.10; 3826.00.21; 3826.00.22; 3826.00.30

Source: Post analysis based on cited legal documents.

3. SAF and Jet Fuel Market

Currently, Vietnam does not produce sustainable aviation fuel (SAF), an alternative to fossil aviation fuel. Additionally, Vietnam currently does not mandate SAF consumption. In late 2025, the Deputy Prime Minister directed two major petroleum product distributors, PetroVietnam and Petrolimex, to begin developing a SAF development program through 2030 with a vision to 2050.

MOIT also issued Directive 13 on August 11, 2025, on the development and use of SAF for aviation transportation for environmental protection and international regulatory compliance. The Directive assigned responsibilities to MOIT departments, related associations, and industries to advance domestic SAF production from agricultural feedstocks including bagasse, cassava peels, rice husks and straw, corn, molasses, roundwood, used cooking oil (UCO), and other biomasses.

Prime Minister Decision 876, dated July 22, 2022, on the Action Program on Green Energy Transition and Reduction of Carbon and Methane Emissions of the Transportation Sector indicated that Vietnam would begin using SAF starting in 2035. During 2022–2030, Vietnam will focus on research and development of SAF technology, policy, and data.

Table 5: Oil Products Consumption in Million Liters

Type	2022	2023	2024	2025
Diesel	10,614	10,873	11,890	12,468
Aviation fuel	1,517	1,594	1,642	1,468

Source: International Energy Agency (IEA)

Financial Supports for Biofuel Producers

Vietnam offers various investment incentives for renewable energy projects, though biofuel-specific support remains limited compared to other renewable energy sectors such as solar and wind.

Corporate Income Tax (CIT) Reductions: Renewable energy projects, including biofuel production facilities, may qualify for preferential CIT rates of 10 percent versus the standard 20 percent, as well as tax holidays typically consisting of two to four years of exemption plus a 50 percent reduction for subsequent years.

Import Duty Exemptions: Equipment and machinery for biofuel production not manufactured domestically may qualify for import duty exemptions or reductions.

Land Use Incentives: Biofuel projects in designated economic zones or rural development areas may receive land rent reductions or exemptions.

Policy/Programs Impacting Rate of Growth in the Fuel Pool

Fuel Pricing System

Vietnam's fuel pricing system significantly impacts biofuel competitiveness. Petroleum product prices are regulated through a price stabilization fund mechanism managed by MOF, with prices adjusted periodically based on international crude oil prices and domestic market conditions.

Key aspects of Vietnam's fuel pricing policy relevant to biofuels:

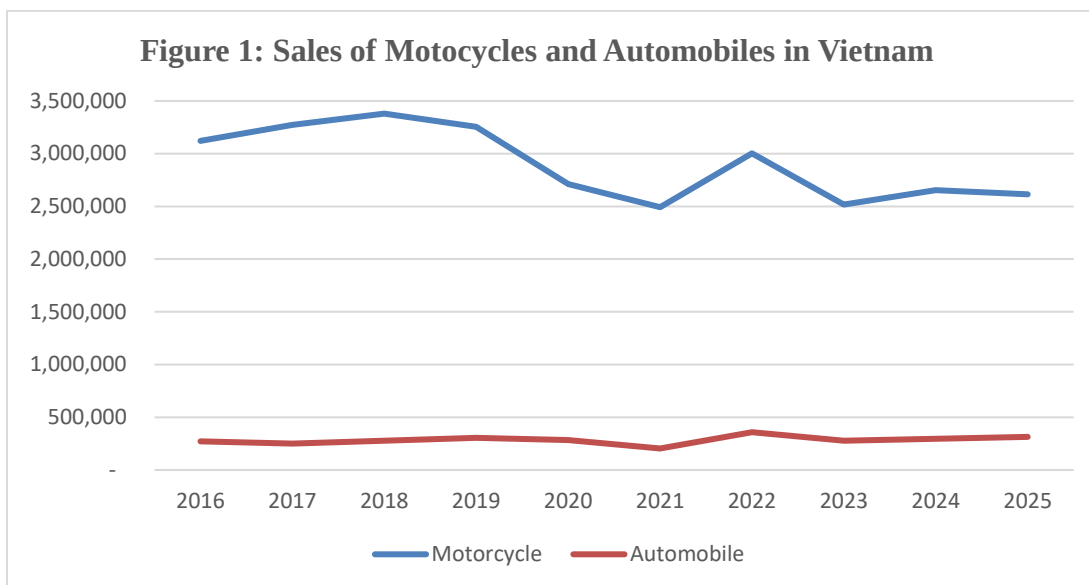
Price Stabilization Fund: The fund collects contributions when international prices are low and provides subsidies when prices spike, aiming to moderate price volatility for consumers.

Electric Vehicle and Two-Wheelers

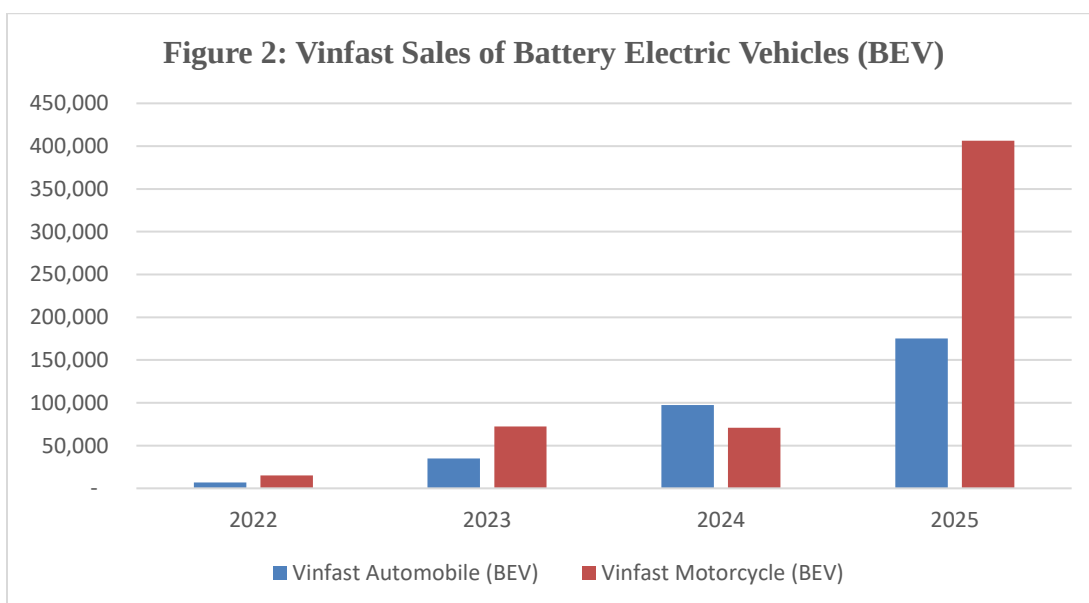
Prime Minister Decision 876, dated July 22, 2022, on the Action Program on Green Energy Transition and Reduction of Carbon and Methane Emissions of the Transportation Sector is one of several important documents issued by the Government and the Prime Minister toward achieving net zero. One of the Action Program's strategies is to promote the production, assembly, import, and transition to electric vehicles and expand blending and use of E5 for all road vehicles by 2030. Vietnam encourages the development of electric vehicles through various tax and fee incentive policies and subsidies for public transport. VinFast, a local brand, produces electric automobiles and motorcycles. Hybrid electric automobiles are also available in Vietnam's market.

According to Vietnam Register, in 2025 there were 5.47 million active automobiles registered for technical and safety inspection as required by law, of which approximately 650,000 were newly registered. The transition to battery electric vehicles (BEV) and hybrid vehicles (HV) has begun in Vietnam over the past two years. VAMA reported HV sales of 14,171 units in 2025, an increase of 44 percent compared to 2024. Approximately 400,000 VinFast battery electric automobiles were sold in 2025, though mostly for the Green SM Taxi fleet. Sales of battery electric automobiles and motorcycles/scooters have increased over the past two years, driven by government tax and fee incentives (Table 7) and fuel savings. VinFast, a local battery electric vehicle manufacturer, has invested in charging station infrastructure across the country to gradually address consumer concerns.

According to local media, Vietnam has more than 75 million active motorcycles and mopeds. Sales of fuel-powered motorcycles and mopeds have remained flat in recent years due to the growing trend toward electric alternatives (Figures 2 and 3).



Source: Vietnam Association of Motorcycle Manufacturer (VAMM) and Vietnam Automobile Manufacturers' Association



Source: Vinfast website and local media.

According to one fuel distributor in the South, the increase in battery electric automobiles and motorcycles/scooters has reduced fuel sales in urban areas, with little impact in rural areas due to a lack of electric charging stations. Petrolimex, the largest local fuel distributor, has prepared an outlook for expected decreasing sales due to projected transition into green transportation in urban areas.

Table 6: Tax and Fees for Electric Vehicles and their Materials and Parts

Item	Fuel Use Automobile	Battery Electric Automobile	Hybrid Automobile	Legal document
Special Consumption tax	10-35%	1-3% till March 1, 2027 4-11% after January 1, 2031	Equal to 70% of the fuel use automobile	Law 66/2025/QH15 Law 09/2026/QH16
Registration Fee	10-12% of the vehicle's value	Zero percent till December 31, 2030	10-12% of the vehicle's value	Government Decree 10/2022 and Decree 51/2025, Decree 202/2026

Source: Post analysis based on cited legal documents.

Vehicle fleet efficiency and emissions policy

Table 7: Automobile Emissions Limit Policy

Emissions	Automobiles produced					Legal document
	Before 1999	1999-2016	2017-2021*	after 2022**	after 2032	
	Level 1	Level 2	Level 3	Level 4	Level 5	
CO (%)	4.5	3.5	3.0	0.5	0.3	Prime Minister Decision 43/2025 and National Technical Regulation 85:2025
HC (ppm)						
04-stroke engine	1,200	800	600	300	150	
02-stroke engine	7,800	7,800	7,800	7,800	7,800	
Special engine	3,300	3,300	3,300	3,300	3,300	

Note: *Automobiles produced in 2017-2022 used in Hanoi and HCMC must comply with level 4

** Automobiles produced after 2022 used in Hanoi and HCMC must comply with level 5 on January 1, 2028

From January 1, 2029, all automobiles used in Hanoi and HCM must comply with level 2 and above.

Automobile Emissions Policy entered into force on March 1, 2026, according to PM Decision 43/2025. With new regulations on increased thresholds of automobile and motorcycles emissions, Vietnam's policy aims to achieve low emissions from transportation and encourage battery vehicles to address air quality pollution.

Table 8: Motorcycles and Mopeds Emissions Limit Policy

Emissions	Level 1	Level 2	Level 3	Level 4	Legal document
CO (%)	4.5	4.5	3.5	2.0	National Technical Regulation 99:2025
HC (ppm)					
04-stroke engine	1,500	1,200	1,100	1,000	
02-stroke engine	10,000	7,800	2,000	2,000	

Note: CO: Carbon Monoxide and HC: Hydrocarbon

According to Prime Minister Decision 13, issued April 2, 2026, and entering into force May 16, 2026, emission levels have been established for motorcycles and mopeds as defined under Law 36/2024/QH15 on Road Traffic Order and Safety.

- **Level 1:** Motorcycles produced and imported before 2008; mopeds produced and imported before 2016.
- **Level 2:** Motorcycles produced and imported during 2008–2016; mopeds produced and imported during 2017–June 30, 2027.
- **Level 3:** Motorcycles produced and imported during 2017–June 30, 2026.
- **Level 4:** Motorcycles produced and imported from July 1, 2027; mopeds produced and imported from July 1, 2027.

From January 1, 2028, motorcycles and mopeds used in Hanoi and Ho Chi Minh City must comply with Level 2 emission standards. Motorcycles and mopeds used in low-carbon areas in Hanoi must comply with Law 39/2024/QH15 in the capital.

Tax on fuels

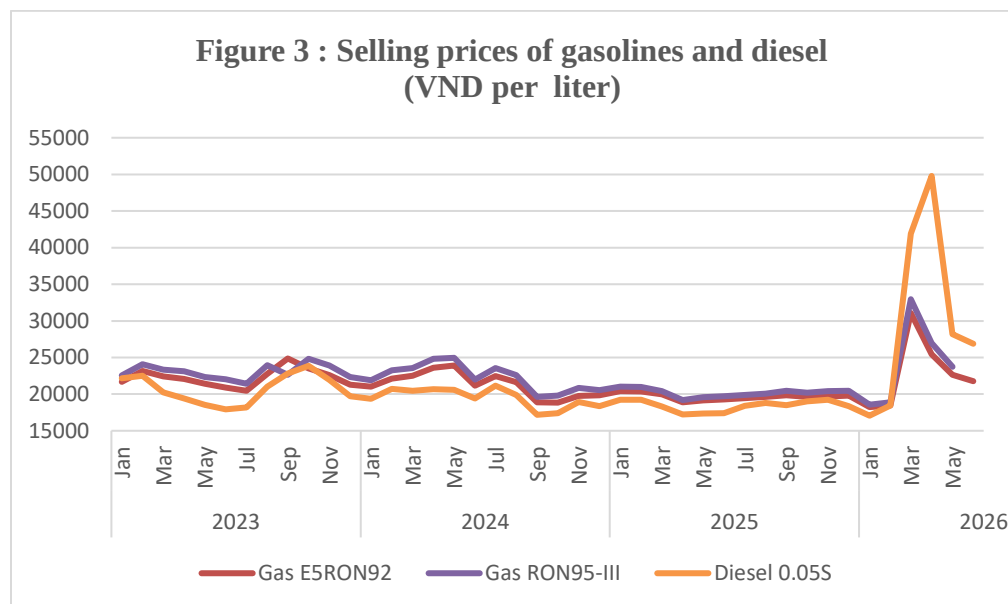
Table 9: Taxes on Fuels

<u>Tax</u>	<u>E5</u> <u>gasoline</u>	<u>E10</u> <u>gasoline</u>	<u>E100</u>	<u>Fossil</u> <u>gasoline</u>	Legal document
Special Consumption tariff rate (SCT)	8%	7%	No	10%	Law 66/2025/QH15
Environmental protection tax	VND 1,900/a liter	VND 1,800/a liter	No	VND 2,000/a liter	Law 57/2010/QH12 Resolution 109/2025/UBTVQH15
MFN Import tariff rate			5%	10%	Government Decree 26 and 73
Value-added tariff rate	8-10%	8-10%	8-10%	8-10%	Law 149/2025/QH15

Source: Post analysis based on cited legal documents.

Vietnam suspended SCT tariff for all gasolines (including E5, E10) to zero percent from April 16 to June 30, 2026, according to National Assembly Resolution 19/2026/QH16 issued on April 12.

Vietnam suspended the environmental protection, MFN import and valued-added tariff rates to zero percent for all gasolines through September 30, 2026, according to National Assembly Resolution 34/2026/QH16 issued on June 30. This reduction lowers selling prices to benefit consumers. Gasoline and diesel prices increased by 50 percent and 100 percent, respectively, due to the impact of Middle East conflicts on global fuel prices (Figure 3).



Source: MOIT official regular announcement on petroleum products prices

Note: Post used the prices for the first week of the month. Currently, fuel prices in Vietnam are adjusted weekly.

Large-scale, public transport investment programs

Hanoi and Ho Chi Minh City are on an investment pathway to build metro systems powered by electricity. Currently, Hanoi has two active metro lines while Ho Chi Minh City has one. The master plan calls for 14 metro lines in Hanoi and 10 lines in Ho Chi Minh City by 2045.

Vietnam plans to replace fuel-powered public buses with electric ones, starting in Hanoi and Ho Chi Minh City, with the goal of transitioning entirely to battery electric buses by 2030. Hanoi has 154 public bus lines while Ho Chi Minh City has more than 200. Currently, approximately 20 bus lines in Hanoi and 10 lines in Ho Chi Minh City are operating battery electric buses.

The Vietnam National Assembly issued Resolution 172/2024/QH15 in 2024 to approve investment in a high-speed North-South Railway spanning 1,541 kilometers at an average speed of 350 kilometers per hour. This investment program has not yet commenced.

On April 12, Vingroup launched a groundbreaking event for an investment project in a high-speed railway from Hanoi to Ha Long, covering 120 kilometers. This project is planned for completion in 2028 and will reduce travel time from Hanoi to Ha Long from two hours to 25 minutes.

Hanoi, as the capital of Vietnam, plans to establish a low-emission zone that would ban and/or restrict fuel-powered vehicles. The boundaries of the low-emission zone are Ring Roads 1, 2, or 3. Post will continue to monitor this development.

C. Environmental Sustainability and Certification

Government Decree 06, issued January 7, 2022, and Decree 119, issued June 9, 2025, require facilities that meet one of the following criteria to implement greenhouse gas (GHG) inventory.

- Power generation facilities and industrial processors/manufacturers have annual energy consumption above 1,000 tons of Oil Equivalent (TOE).
- Transportation companies have annual energy consumption above 1,000 tons of Oil Equivalent (TOE).
- Office buildings have annual energy consumption above 1,000 tons of Oil Equivalent (TOE)
- Solid waste treatment facilities have annual capacity above 65,000 tons

The list of facilities that must do GHG inventory is available in the Prime Minister’s Decision 13 issued in August 2024. One ethanol producer in the south is in the list with total estimated annual energy consumption above 1,350 TOE.

D. Trade Policy, Import Duties/Licenses

Biofuel trade is open to all businesses. Import procedures and documentation requirements for E100 are regulated under Circular 38/2015, issued March 25, 2015, and amended by Circular 39/2018, issued April 20, 2018. Quality inspection, regulated under Decision 2711/QD-BKHCN issued December 30, 2022, is a required step for customs clearance.

On March 31, 2025, the Government of Vietnam issued Decree 73/2025/ND-CP, reducing the Most-Favored-Nation (MFN) import tariff rates on ethanol from 10 to 5 percent. Please go to [GAIN Report VM2025-0014](#) for details. This is the third time Vietnam reduced MFN import tariff rate for ethanol.

Table 10: MFN Import Tariff Rate for Ethanol

HS code - Ethanol	MFN tariff before July 10, 2020 (percent)	MFN tariff under Decree 57/2020 - Entered into force on July 10, 2020 (percent)	MFN tariff under Decree 26/2023 entered into force July 15, 2023 (percent)	MFN tariff under Decree 73/2025 entered into force March 31, 2025 (percent)
2207.20.11	17	15	10	5
2207.20.19	20	15	10	5

Table 11: Import Tariff Rates on Biofuels

Product description	HS code	MFN Import tariff rate (%)		ACFTA*	VKFTA*	ASEAN
		USA	Brazil	China	Korea	
Denatured ethanol	22.07.20.11; 2207.20.19; 2207.20.90	5	5	0	0	0
Udenatured ethanol	2207.10.00	40	40	0	0	0
Biodiesel blended fuel	2710.20.00	0	0	20	0	0
B100	3826.00.10;	0	0	0	0	0

	3826.00.21; 3826.00.22; 3826.00.30					
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Source: Post analysis based on cited legal documents.
**Asean-China Free Trade Agreement and Vietnam-Korea Free Trade Agreement*

SECTION III: ETHANOL

Vietnam's fuel ethanol sector in 2026 continues to face structural challenges that limit its domestic supply. Despite the E5 mandate being in place since 2018 and the E10 mandate taking effect in June 2026, domestic production remains insufficient due to the slow resumption of existing ethanol plants, necessitating continued reliance on imports.

The data table below presents the views of the author and does not reflect the official views of the U.S. Department of Agriculture (USDA). Official government statistics on biofuels are not available. This report is based on analytical assessments, not official data.

Table 12: Vietnamese Ethanol Used as Fuel and Other Industrial Chemicals (in million liters)

Ethanol Used as Fuel and Other Industrial Chemicals (Million Liters)									
Calendar Year	2018	2019	2020	2021	2022	2023	2024	2025	2026f
Beginning Stocks	3	5	3	7	6	17	10	11	8
Fuel Begin Stocks	3	3	2	5	5	9	5	7	2
Production	140	140	120	125	130	125	90	100	210
Fuel Production	105	110	85	80	85	95	60	65	170
Imports	45	24	57	39	35	19	45	11	490
Fuel Imports	44	23	53	38	33	18	45	11	490
Exports	13	11	23	27	14	11	2	2	1
Fuel Exports	9	7	10	8	6	9	1	1	0
Consumption	170	155	150	138	140	140	132	112	685
Fuel Consumption	140	127	125	110	108	108	102	80	653
Ending Stocks	5	3	7	6	17	10	11	8	22
Fuel Ending Stocks	3	2	5	5	9	5	7	2	9
Refineries Producing Fuel Ethanol (Million Liters)									
Number of Refineries	2	2	2	2	2	2	2	3	4
Nameplate Capacity	170	170	170	170	170	170	170	220	320
Capacity Use (%)	61.8%	64.7%	50.0%	47.1%	50.0%	55.9%	35.3%	29.5%	53.1%
Co-product Production (1,000 MT)									
DDGS	50	54	42	40	42	47	30	32	84
Feedstock Use for Fuel Ethanol (1,000 MT)									
Cassava	250	125	41	38	41	45	29	47	122
Corn		144	170	160	170	190	120	114	298
Market Penetration (Million Liters)									
Fuel Ethanol Use	140	127	125	110	108	108	102	80	653
Gasoline Pool 1/	8,882	9,104	7,721	7,482	9,031	9,481	10,049	10,537	10,812
Blend Rate (%)	1.6%	1.4%	1.6%	1.5%	1.2%	1.1%	1.0%	0.8%	6.0%

Source: IEA, derived Vietnam trade via Trade Data Monitor, Post calculation and forecasts, domestic ethanol producers

A. Consumption

Post forecasts fuel ethanol consumption in 2026 will increase to 653 million liters due to the mandatory E10 gasoline requirement nationwide starting June 1 and the continued availability of E5 gasoline in the market. Fossil gasoline, at all grades, is no longer permitted in the market.

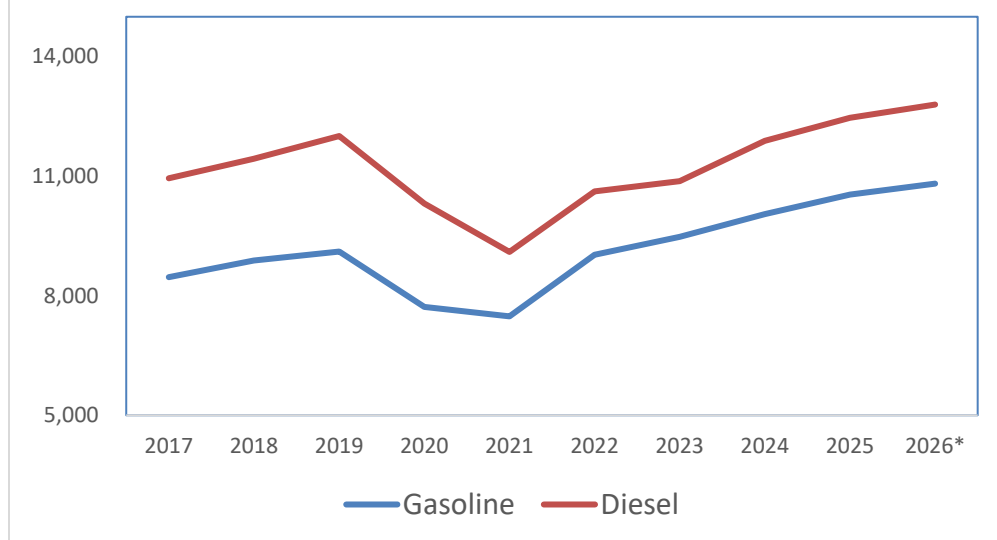
According to local ethanol producers, non-fuel ethanol consumption is used for beverages, pharmacies, chemicals and electronics industries. These are constant and stable markets. Dai Viet Plant which is fully active to produce purely non-fuel ethanol.

Total fuel ethanol consumption in Vietnam for 2025 is estimated at 80 million liters, reflecting the volume needed to meet E5 gasoline demand from the country's gasoline pool. E5 gasoline consumption declined due to low consumer acceptance, reduced E5 blending capacity, and fewer E5 pumps at the station network of one of the largest petroleum companies. Total gasoline consumption in Vietnam has grown steadily, driven by increasing vehicle ownership (figure 4).

In late 2025 and early 2026, consumer acceptance of ethanol-blended gasoline has gradually improved, though some resistance persists, particularly regarding concerns about engine compatibility and fuel efficiency. Thanks to Government of Vietnam, particularly Ministry of Industry and Trade, which provided several education and communication campaigns on biofuel benefits. Public education campaigns and the absence of reported technical problems have helped increase acceptance. Ethanol consumption reduced due to reduced E5 blending capacity, and fewer E5 pumps at the gas station network.

MOIT reported that Vietnam gasoline consumption during June 1 to 28, 2026, was 980 million liters of which 94.29 percent are E10. Local media reported consumers are not easy to find pump stations offering E5.

**Figure 4: Growth in Gasoline and Diesel Pool in Vietnam
(Million Liters)**



Note: 2026 is a forecast

Source: International Energy Agency (IEA)

B. Production

Post forecasts Vietnam's ethanol production will increase to 210 million liters in 2026 due to the resumption of the Dung Quat ethanol plant and increased production at existing ethanol plants to meet growing domestic demand.

Vietnam has seven ethanol plants but the first four below are currently operating to produce ethanol.

1. Tung Lam Ethanol Plant (Dong Nai Province)

Started: 2009

Feedstock: corn kernels, cassava

Nameplate Capacity: ~70 million liters per year (~70,000 cubic meters per year)

Actual annual production: 80% ~ 60 million liters

Outputs: fuel ethanol, beverage, and industrial uses (cosmetics, electricity)

Status: relatively consistent operation, close to major fuel demand center

2. Dung Quat Biofuel Plant / BSR (Quang Ngai Province)

Resumed: February 2026

Feedstock: corn kernels, cassava

Capacity: ~100 million liters per year (~100,000 cubic meters per year)

Actual annual production: 60% ~ 60 million liters

Output: purely fuel ethanol

Status: strategic facility linked to PetroVietnam/BSR system for blending to support E10

3. Quang Nam Ethanol Plant (Quang Nam Province)

Started: 2009

Feedstock: corn kernels, cassava

Capacity: ~100 million liters per year (~100,000 cubic meters per year)

Actual annual production: 70% ~ 70 million liters

Output: fuel ethanol and industrial uses (electricity, pharmacy)

Status: operating but historically underutilized depending on market conditions

4. Dai Viet Ethanol Plant (Dak Nong Province)

Resumed: January 2025

Feedstock: cassava

Nameplate Capacity: ~50 million liters per year (~50,000 cubic meters per year)

Actual annual production: 60% ~ 30 million liters

Output: for industrial uses (electricity, pharmacy, chemicals)

Status: intermittent operation depending on economics and feedstock costs

The other three ethanol plants below are not actively producing ethanol due to the closure of the Binh Phuoc and Dak To plants, and the unfinished construction of the Binh Phuoc plant. The Government of Vietnam encourages these plants to be renovated to produce ethanol for domestic consumption.

5. Binh Phuoc Ethanol Plant (Binh Phuoc Province)

6. Tam Nong Ethanol Plant (Phu Tho Province)

7. Dak To Ethanol Plant (Kon Tum)

Feedstock Availability and Pricing

Cassava prices in Vietnam are highly volatile, influenced by weather conditions, competing export demand (primarily to China for starch production), and domestic food and feed requirements. According to an industry contact, cassava is also sourced from neighboring countries including Laos and Cambodia. When cassava prices spike, ethanol production becomes economically unviable, forcing plants to reduce output or temporarily shut down. According to ethanol producers, sourcing dried cassava chips in the first quarter of the year, during the dry season, is efficient and effective for ethanol production, after which they prefer using corn for the remainder of the year. Fresh cassava is no longer used for ethanol production due to its short shelf life and high-water consumption.

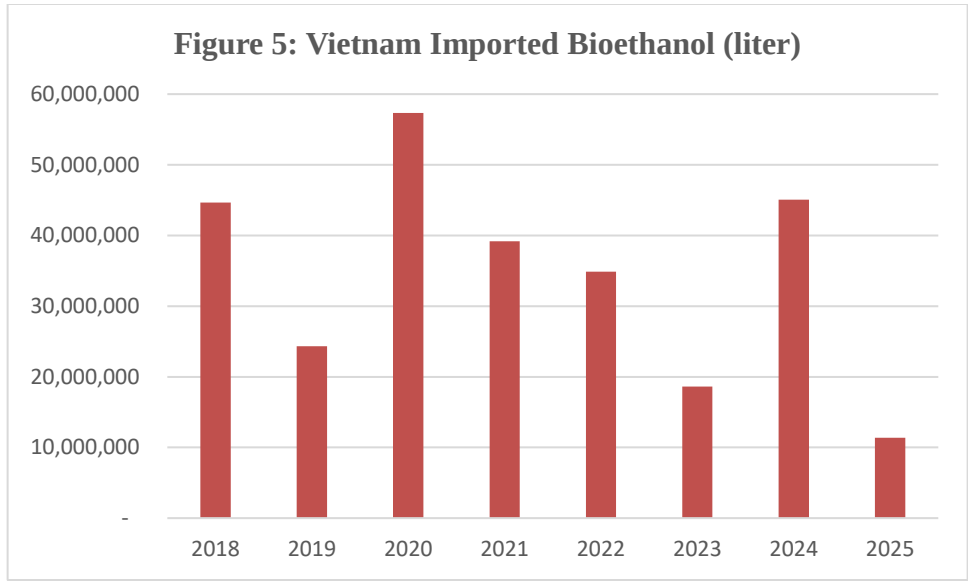
Corn kernels: Local ethanol plants are sourcing South American and U.S. corn from traders. Local corn volume is not sufficient to supply the demand.

Vietnam's ethanol plants utilize traditional conversion technologies that are less efficient than facilities in Brazil and the United States. Higher production costs per liter may reduce competitiveness against imports.

Seasonal Variations: Feedstock availability varies seasonally, particularly for cassava, leading to uneven production throughout the year.

Molasses: Vietnam does not currently use molasses for ethanol production. Post projects potential investment in future ethanol plants utilizing this feedstock.

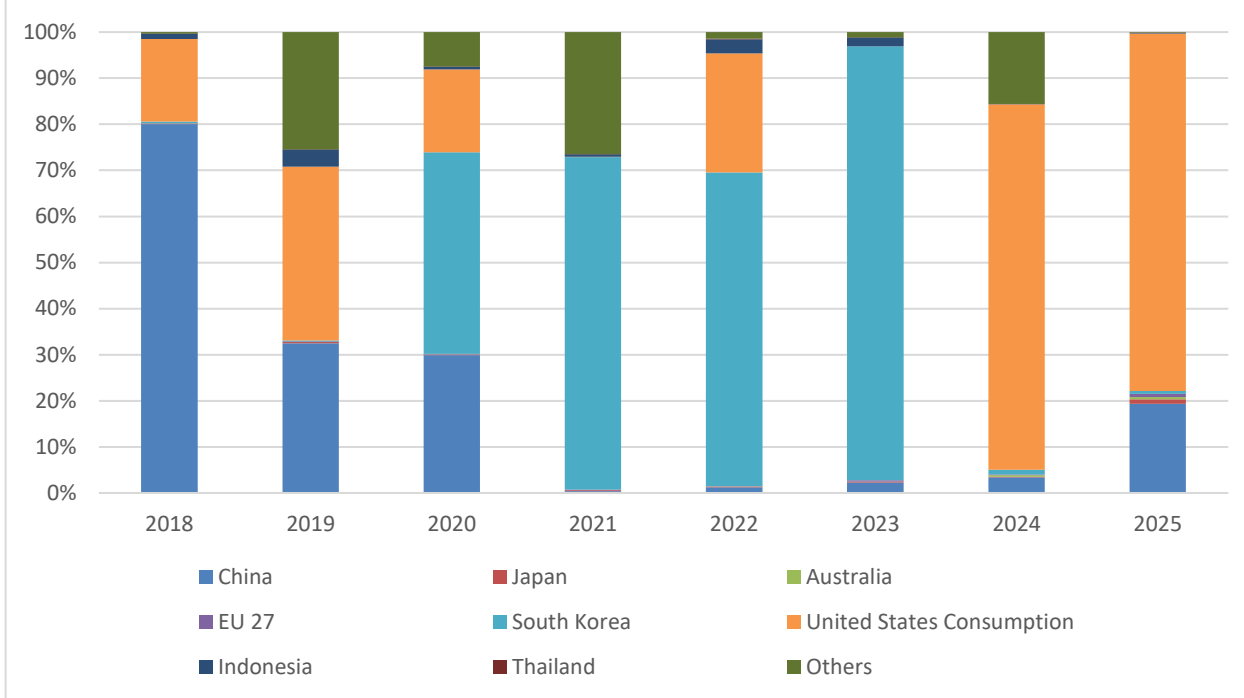
C. Trade



Source: Trade Data Monitor¹

¹ Vietnam is not a reported country in TDM. Throughout this report, every time TDM is noted, Vietnam exports refer to reported global imports from Vietnam and Vietnam imports refers to global exports reported to Vietnam.

Figure 6: Vietnam's Ethanol Import Marketshare



Source: Trade Monitor Data

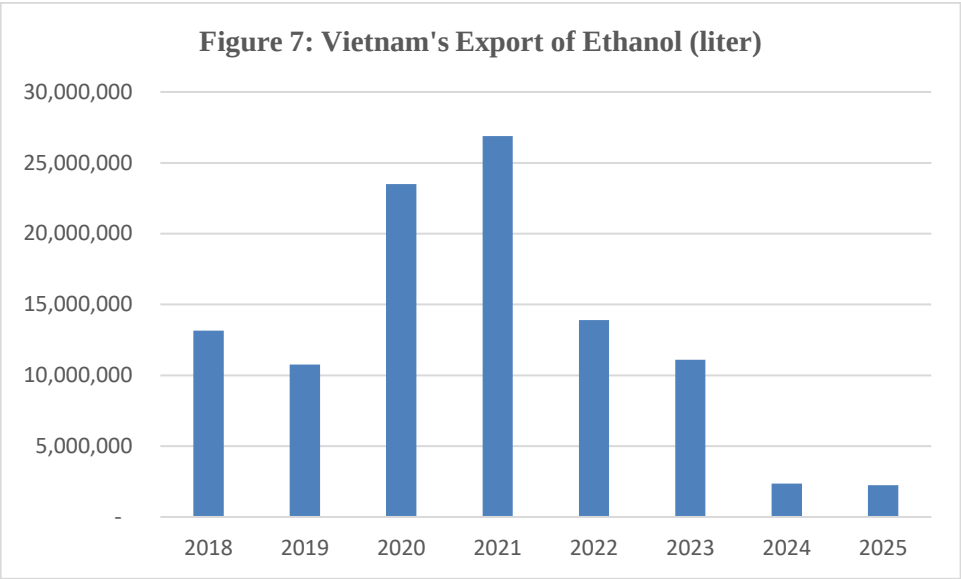
Ethanol Imports

Post forecasts Vietnam's fuel ethanol imports will increase to 490 million liters in 2026 due to the mandatory E10 and E5 gasoline requirements nationwide starting June 1.

Vietnam's fuel ethanol imports have mostly trended downward since 2021 due to decreased market share of E5 gasoline. Over the past two years, the United States has been the main supplier.

Ethanol Exports

Vietnam does not export significant volumes of fuel ethanol, as domestic production is insufficient to meet demand. Vietnam exports small volumes of non-fuel ethanol.



Source: Trade Monitor Data

SECTION IV: BIODIESEL

Vietnam's biodiesel sector remains in early-stage development, with minimal commercial production and no nationwide blending mandate. Unlike ethanol, which benefits from the E5 and E10 mandates and established production facilities, biodiesel faces more fundamental challenges related to feedstock availability and policy support.

A. Consumption

Vietnam's biodiesel sector consumption remains minimal, with pilot programs in select provinces — Ho Chi Minh City, Binh Duong, and Dong Nai — testing B5 blends in buses, trucks, and government vehicles, as well as use in industrial applications such as generators and agricultural machinery.

Diesel Fuel Pool: Vietnam's diesel consumption has grown steadily, driven by freight transport, construction, agriculture, and industrial activity. Total diesel consumption in 2025 is estimated at 11.3 billion liters.

The absence of a national biodiesel mandate means biodiesel penetration remains negligible. Industry stakeholders consistently cite the need for a clear mandate, to create demand certainty and justify investment in production capacity.

B. Production

Biodiesel production in Vietnam is limited to small-scale facilities and pilot projects.

C. Trade

Biodiesel Imports: Vietnam's biodiesel imports and exports remain negligible. Imports are primarily for pilot programs or specialized industrial applications.

SECTION V: ADVANCED BIOFUELS

Vietnam's advanced biofuels sector, particularly SAF, is in early-stage development but showing increasing momentum driven by international aviation industry commitments and regional investment trends.

Sustainable Aviation Fuels (SAF)

Vietnam has no commercial-scale SAF production facilities and no official SAF mandate. However, at the start of 2026, Vietnam began participating on the voluntary phase of the UN Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), which is tasked with reducing carbon dioxide emissions from the aviation sector. Additionally, private companies in Vietnam made several voluntary actions to boost future SAF consumption.

Vietnam has two major domestic airlines. Vietnam Airlines conducted its first SAF trial in May 2024, when an Airbus A321 used a SAF-blended fuel for a passenger flight from Singapore, followed by domestic pilot flights in June 2025 from Da Nang to Hanoi and Ho Chi Minh City. VietJet Air used imported SAF for two international flights from Ho Chi Minh City to Melbourne and Seoul on October 17, 2024.

Vietnam Airlines, the country's flag carrier, has announced commitments aligned with IATA's net-zero 2050 target. The airline has:

- Signed a memorandum of understanding (MOUs) with international SAF suppliers for future offtake
- Participated in regional SAF working groups and industry forums
- Conducted initial SAF test flights using imported SAF blends on select international routes
- Announced targets to incorporate SAF into its fuel mix by 2026–2027, starting with international long-haul routes

Petrovietnam, Vietnam's state-owned oil and gas company, expressed interest in SAF production as part of its energy transition strategy:

- Feasibility studies underway for potential biorefinery projects
- Exploring partnerships with international technology providers
- Assessing feedstock availability, particularly UCO and agricultural residues
- Potential sites identified near existing refinery infrastructure at Dung Quat and Nghi Son

Feedstock Considerations: Potential SAF feedstocks in Vietnam include:

- UCO: Current collection rates are low.
- Animal and fish fats: Limited availability from domestic livestock and fish processing
- Agricultural residues: Abundant, including rice straw, corn stover, and sugarcane bagasse, but requiring advanced conversion technologies

Policy and Regulatory Framework: Vietnam does not have specific SAF policies or mandates.

Appendix A: Statistical Information

Feedstock to Biofuel Conversion Rates

Source: local ethanol producers

Ethanol:

Cassava (dried chips)	1MT = 418 liters
Corn kernels	1MT = 400 liters

Ethanol feedstocks to Co-products

Cassava (dried chips)	1MT = 200 kg DDGS
Corn kernels	1MT = 200 kg DDGS

Trade Data

Trade data are derived from Trade Data Monitor (TDM). Because Vietnam does not report trade data to TDM, Vietnam's trade statistics are estimated using partner-country data. Vietnam imports are based on exports to Vietnam reported by partner countries, while Vietnam exports are based on imports from Vietnam reported by partner countries.

For ethanol, trade is tracked using HS Codes 2207.10 (undenatured ethyl alcohol) and 2207.20 (denatured ethyl alcohol). Because customs classifications do not distinguish fuel from non-fuel ethanol, Post applies the following assumptions:

Imports under HS Code 2207.20 are treated as fuel ethanol.

Imports from Thailand and China under HS Code 2207.10 are also treated as fuel ethanol, based on established trade patterns.

All other imports under HS Code 2207.10 are treated as non-fuel ethanol.

Ethanol exports under HS Code 2207.20 are treated as fuel ethanol.

For biodiesel, trade is calculated using HS Code 3826.00, which is assumed to represent B100 biodiesel, and HS Code 2710.20, which is assumed to represent B5 biodiesel blends. Volumes reported under HS Code 2710.20 are converted to their B100-equivalent volume for reporting purposes.

Other Data Sources

- International Energy Agency for gasoline and diesel pool
- Ethanol production from local producers

Attachments:

No Attachments.